

CAMBRIDGE
INTERNATIONAL EXAMINATIONS

June 2003

GCE ADVANCED SUBSIDIARY LEVEL AND ADVANCED LEVEL

MARK SCHEME

MAXIMUM MARK: 40

SYLLABUS/COMPONENT: 9702/01

PHYSICS
Paper 1 (Multiple Choice (AS))



UNIVERSITY *of* CAMBRIDGE
Local Examinations Syndicate

Page 1	Mark Scheme	Syllabus	Paper
	A/AS LEVEL EXAMINATIONS - JUNE 2003	9702	01

<i>Question Number</i>	<i>Key</i>	<i>Question Number</i>	<i>Key</i>
1	B	21	B
2	B	22	D
3	B	23	B
4	A	24	D
5	C	25	C
6	B	26	B
7	C	27	C
8	C	28	C
9	D	29	B
10	D	30	C
11	B	31	A
12	A	32	B
13	D	33	B
14	B	34	B
15	A	35	C
16	C	36	D
17	C	37	B
18	D	38	C
19	B	39	B
20	A	40	D

CAMBRIDGE

INTERNATIONAL EXAMINATIONS

June 2003

GCE ADVANCED SUBSIDIARY LEVEL AND ADVANCED LEVEL

MARK SCHEME

MAXIMUM MARK: 60

SYLLABUS/COMPONENT: 9702/02

PHYSICS

Paper 2 (Structured Questions (AS))



UNIVERSITY of CAMBRIDGE
Local Examinations Syndicate

Page 1	Mark Scheme	Syllabus	Paper
	A/AS LEVEL EXAMINATIONS - JUNE 2003	9702	02

Categorisation of marks

The marking scheme categorises marks on the *MACB* scheme.

B marks: These are awarded as independent marks, which do not depend on other marks. For a B-mark to be scored, the point to which it refers must be seen specifically in the candidate's answer.

M marks: These are method marks upon which A-marks (accuracy marks) later depend. For an M-mark to be scored, the point to which it refers must be seen in the candidate's answer. If a candidate fails to score a particular M-mark, then none of the dependent A-marks can be scored.

C marks: These are compensatory method marks which can be scored even if the points to which they refer are not written down by the candidate, providing subsequent working gives evidence that they must have known it. For example, if an equation carries a C-mark and the candidate does not write down the actual equation but does correct working which shows he/she knew the equation, then the C-mark is awarded.

A marks: These are accuracy or answer marks which either depend on an M-mark, or allow a C-mark to be scored.

Conventions within the marking scheme

BRACKETS

Where brackets are shown in the marking scheme, the candidate is not required to give the bracketed information in order to earn the available marks.

UNDERLINING

In the marking scheme, underlining indicates information that is essential for marks to be awarded.

Page 2	Mark Scheme	Syllabus	Paper
	A/AS LEVEL EXAMINATIONS - JUNE 2003	9702	02

- 1 kg m⁻³ B1
frequency or count rate or activity or decay constant B1
NC⁻¹ or V m⁻¹ or kg m s⁻² C⁻¹ etc. B1
momentum or impulse..... B1 [4]
(Allow solidus notation and non SI units)
- 2 (a) (i) distance from a (fixed) point.....M1
in a specified direction A1
(Allow 1 mark for 'distance in a given direction')
- (ii) (displacement from start is zero if) car at its starting position..... B1 [3]
- (b) (i)1 $v^2 = u^2 + 2as$
 $28^2 = 2 \times a \times 450$ (use of component of 450 scores no marks)..... C1
 $a = 0.87 \text{ m s}^{-2}$ A1 [2]
(-1 for 1 sig. fig. but once only in the question)
- (i)2 $v = u + at$ or any appropriate equation
 $28 = 0.87t$ or appropriate substitution..... C1
 $t = 32 \text{ s}$ A1 [2]
- (i)3 $E_k = \frac{1}{2}mv^2$ C1
 $= \frac{1}{2} \times 800 \times 28^2$
 $= 3.14 \times 10^5 \text{ J}$ A1 [2]
- (i)4 $E_p = mgh$ C1
 $= 800 \times 9.8 \times 450 \sin 5$ C1
 $= 3.07 \times 10^5 \text{ J}$ A1 [3]
- (ii) power = energy/time C1
 $= (6.21 \times 10^5)/32.2$ C1
 $= 1.93 \times 10^4 \text{ W}$ A1 [3]
(power = Fv with $F = mg \sin \theta$ scores no marks)
- (iii) some work also done against friction forces.....M1
location of frictional forces identified A1 [2]
(allow reasonable alternatives)
- 3 (a) (i) ductile B1
- (ii)1 L shown at end of straight line B1
- (ii)2 reciprocal of gradient of straight line region B1 [3]
- (b) (i)1 circumference = $3\pi \text{ cm}$ or arc = $r\theta$ C1
extension = $(6.5/360) \times 3\pi$ M1
 $= 0.17 \text{ cm}$ A0
- (i)2 strain = extension/length..... C1
 $= 0.17/250$
 $= 6.8 \times 10^{-4}$ A1 [4]
- (ii) stress = force/area C1
 $= (6.0 \times 9.8)/(7.9 \times 10^{-7})$ C1
 $= 7.44 \times 10^7 \text{ Pa}$ A1 [3]

Page 3	Mark Scheme	Syllabus	Paper
	A/AS LEVEL EXAMINATIONS - JUNE 2003	9702	02

- (iii) Young modulus = stress/strain..... C1
 $= (7.44 \times 10^7)/(6.8 \times 10^{-4})$
 $= 1.1 \times 10^{11} \text{ Pa}$ A1 [2]
- (iv) remove extra load and see if pointer returns to original position or wire returns to original length B1 [1]
- 4 (a) e.g. both transverse/longitudinal/same type meet at a point, same direction of polarisation, etc.....1 each, max 3 B3 [3]
(allow 1 mark for any condition for observable interference)
- (b) (i)1 allow 0.3 mm $\rightarrow$ 3 mm..... B1
(ii)2 $\lambda = ax/D$ (allow any subject) B1
(ii)1 separation increased B1
less bright B1
(ii)2 separation increased B1
less bright B1
(ii)3 separation unchanged B1
fringes brighter B1
further detail, i.e quantitative aspect in (ii)1 or (ii)2 B1 [7]
(in (b), do not allow e.c.f. from (b)(i)2)
- 5 (a) (i) resistance = V/I C1
 $= 6.0/(40 \times 10^{-3})$
 $= 150 \Omega$ A1
(no marks for use of gradient)
(ii) at 8.0 V, resistance = $8.0/(50 \times 10^{-3}) = 160 \Omega$ C1
change = 10Ω A1 [4]
- (b) (i) straight line through origin M1
passes through $I = 40 \text{ mA}$, $V = 8.0 \text{ V}$ A1
(ii) current in both must be 40 mA C1
e.m.f. = $8.0 + 6.0 = 14.0 \text{ V}$ A1 [4]
- 6 (a) (i) curve is not smooth, fluctuations, etc B1
(ii) curve is same shape or same half-life, not affected by temperature, etc B1 [2]
- (b) (i) 134..... B1 [1]
(ii) α -particle shown as ${}^4_2\text{He}$ or as ${}^4_2\alpha$ B1
nucleon number of Po shown as 216 B1
proton number of Po shown as 84..... B1 [3]

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INTERNATIONAL EXAMINATIONS

June 2003

GCE ADVANCED SUBSIDIARY LEVEL AND ADVANCED LEVEL

MARK SCHEME

MAXIMUM MARK: 25

SYLLABUS/COMPONENT: 9702/03

**PHYSICS
Paper 3 (Practical (AS))**



UNIVERSITY of CAMBRIDGE
Local Examinations Syndicate

Page 1	Mark Scheme	Syllabus	Paper
	A/AS LEVEL EXAMINATIONS - JUNE 2003	9702	03

- 1 (a) (iv) % uncertainty in θ 2/1/0
Accept $\Delta\theta$ to $\pm 1^\circ \pm 2^\circ$ (1 mark)
Ratio and percentage ideas correct (1 mark)
- (d) (i) Measurements 3/2/1/0
Expect to see at least 6 sets of results (1 mark)
Less than 6 sets does not score this mark
Check a value of T^4 . Underline checked value and tick if correct (1 mark)
Ignore small rounding errors. This mark cannot be awarded if there are no raw times, number of oscillations measured in a fixed time, or the stopwatch has been misread. If there is no record of the number of oscillations then this mark cannot be scored
It may be necessary to refer to page 3 of script for a value of n
Check a value for $\cos\theta$. Underline checked value and tick if correct (1 mark)
Ignore small rounding errors. Expect to see a correct sign
If either incorrect, write in correct value and -1 eeo
Minor help given by Supervisor, -1. Major help, then -2
- (d) (i) Repeated readings 1
For each value of θ there must be at least two values of t
An average value does **not** have to be calculate
- (d) (i) At least 10° between the readings of θ 1
- (d) (i) Quality of results 2/1/0
Judge by scatter of points about Examiner line of best fit
6 reasonable trend plots with little scatter (2 marks)
5 trend plots, or some scatter of plots (1 mark)
Large scatter/no trend/wrong quantities plotted (zero)
- (d) (i) Column headings 1
Check the $1/T^4$ column heading only
Quantity and unit (s^{-4}) must be correct
- (d) (i) Consistency 2/1/0
Apply to raw values of θ **and** t only (one mark each)
Values of θ must all be given to the nearest degree. Do not allow tenths of a degree
Values of t must all be given to the nearest 0.1 s or 0.01 s
Do not apply to average values
- (d) (ii) Justification of number of sf in $\cos\theta$ 1
Answer must relate sf in θ to sf in $\cos\theta$
Do not allow answers in terms of decimal places
Do not allow vague answers that are given in terms of 'raw data'
- (e) (i) Axes 1
Scales must be such that the plotted points occupy at least half the graph grid in both the x and y directions (i.e. 4 x 6 in portrait or 6 x 4 in landscape)
Axes must be labelled with the quantity plotted. Ignore units. Do not allow awkward scales or gaps of more than three large squares between the scale markings

Page 2	Mark Scheme	Syllabus	Paper
	A/AS LEVEL EXAMINATIONS - JUNE 2003	9702	03

- (e) (i) Plotting of points 1
Check a suspect plot. Circle and tick if correct. If incorrect, show correct position with arrow, and -1. Work to half a small square. All observations must be plotted
- (e) (i) Line of best fit 1
There must be a reasonable balance of points about the line of best fit
There must be at least 5 plots on the grid for this mark to be awarded
Do not allow a straight line to be drawn through a distinct curve trend
Allow an acceptable curve through a curved trend of points
- (e) (ii) Determination of gradient 1
Hypotenuse of Δ used must be greater than half the length of the drawn line
Check the read-offs and ratio. Read-offs must be accurate to half a small square
Do not allow this mark if a curve has been drawn
- (e) (ii) y-intercept 1
The value must be read to half a small square
Do not allow this mark if a curve has been drawn
- (f) A = candidate's value of gradient 1
- (f) B = candidate's value of intercept 1
- (f) Unit of A and B **both** correct (s^{-4}) 1
- (g) Measurement of L 1
The value should be in the range $40 \text{ cm} \pm 2 \text{ cm}$. Can be implied in the working
It may be necessary to refer to the Supervisor's Report
- (g) Correct method of working to give a value for g in range 9.0 to 11.0 m s^{-2} 1
A POT error anywhere in the working will not score this mark
- (g) Sf in g 1
Allow 2 or 3 sf only. Apply to any value given
A bald value with no working cannot score this mark
- (g) Unit of g correct (and consistent with other measurements, e.g. L) 1
There must be a numerical value of g for this mark to be scored
A bald value with no working cannot score this mark

25 marks in total

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GCE ADVANCED SUBSIDIARY LEVEL AND ADVANCED LEVEL

MARK SCHEME

MAXIMUM MARK: 60

SYLLABUS/COMPONENT: 9702/04

PHYSICS

Paper 4 (Structured Questions (A2 Core))



UNIVERSITY of CAMBRIDGE
Local Examinations Syndicate

Page 1	Mark Scheme	Syllabus	Paper
	A/AS LEVEL EXAMINATIONS - JUNE 2003	9702	04

Categorisation of marks

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M marks: These are method marks upon which A-marks (accuracy marks) later depend. For an M-mark to be scored, the point to which it refers must be seen in the candidate's answer. If a candidate fails to score a particular M-mark, then none of the dependent A-marks can be scored.

C marks: These are compensatory method marks which can be scored even if the points to which they refer are not written down by the candidate, providing subsequent working gives evidence that they must have known it. For example, if an equation carries a C-mark and the candidate does not write down the actual equation but does correct working which shows he/she knew the equation, then the C-mark is awarded.

A marks: These are accuracy or answer marks which either depend on an M-mark, or allow a C-mark to be scored.

Conventions within the marking scheme

BRACKETS

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UNDERLINING

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Page 2	Mark Scheme	Syllabus	Paper
	A/AS LEVEL EXAMINATIONS - JUNE 2003	9702	04

- 1 (a) work done in bringing/moving unit mass M1
from infinity to the point..... A1 [2]
(use of 1 kg in the definition – max 1/2)
- (b) potential at infinity defined as being zero..... B1
forces are always attractive..... B1
so work got out in moving to point..... B1 [3]
(max potential is at infinity – allow 1/3)
- (c) (i) $\phi = -GM/R$
change = $6.67 \times 10^{-11} \times 6.0 \times 10^{24} \times \{6.4 \times 10^6\}^{-1} - \{1.94 \times 10^7\}^{-1}$ C2
change = $4.19 \times 10^7 \text{ J kg}^{-1}$ (ignore sign) A1
- (ii) $\frac{1}{2}mv^2 = m\Delta\phi$ C1
 $v^2 = 2 \times 4.19 \times 10^7 = 8.38 \times 10^7$
 $v = 9150 \text{ m s}^{-1}$ A1 [5]
- (d) acceleration is not constant..... B1 [1]
- 2 (a)  (-1 for each error or omission) B2 [2]
- (b) heat lost by liquid gold = $0.95m \times 129 \times \Delta T$ C1
heat gained (silver) = $0.05m \times 235 \times (1340 - 300) + 0.05m \times 105\,000$ C1, C1
 $122.5m\Delta T = 17\,470m$
 $\Delta T = 143 \text{ K}$ C1
temperature = $143 + 1340 = 1483 \text{ K}$ A1 [5]
- (c) e.g. thermocouple/resistance thermometer B1 [1]
- 3 (a) f_0 is at natural frequency of spring (system) B1
this is at the driver frequency B1 [2]
(allow 1 mark for recognition that this is resonance)
- (b) line: amplitude less at all frequencies B1
peak flatter B1
peak at f_0 or slightly below f_0 B1 [3]
- (c) (aluminium) sheet cuts the magnetic flux/field B1
(so) currents/e.m.f. induced in the (metal) sheet B1
these currents dissipate energy M1
less energy available for the oscillations A1
so amplitude smaller A0 [4]
(‘current opposes motion of sheet’ scores one of the last two marks)
- 4 (a) field causes forces on the electrons M1
and the nucleus in opposite directions A1
(field causes) electrons (to be) stripped off the atom..... B1 [3]
- (b) (i) $E = Q/4\pi\epsilon_0 r^2$ C1
 $20 \times 10^3 \times 10^2 = Q/(4\pi \times 8.85 \times 10^{-12} \times 0.21^2)$ C1
charge = $9.8 \times 10^{-6} \text{ C}$ A1 [3]

Page 3	Mark Scheme	Syllabus	Paper
	A/AS LEVEL EXAMINATIONS - JUNE 2003	9702	04

- (ii) $V = Q/4\pi\epsilon_0 r$
 $= (9.8 \times 10^{-6})/(4\pi \times 8.85 \times 10^{-12} \times 0.21)$C1
 $= 4.2 \times 10^5 \text{ V}$A1 [2]
- (c) e.g. sphere not smooth, humid air, etcB1 [1]
- 5 (a) centripetal force $= mv^2/r$B1
magnetic force $F = Bqv$B1
(hence) $mv^2/r = Bqv$B1
 $r = mv/Bq$ A0 [3]
- (b) $r_\alpha/r_\beta = (m_\alpha/m_\beta) \times (q_\beta/q_\alpha)$ C1
 $= (4 \times 1.66 \times 10^{-27})/(9.11 \times 10^{-31} \times 2)$
 $= 3.64 \times 10^3$ A2 [3]
- (c) (i) $r_\alpha = (4 \times 1.66 \times 10^{-27} \times 1.5 \times 10^6)/(1.2 \times 10^{-3} \times 2 \times 1.6 \times 10^{-19})$
 $= 25.9 \text{ m}$ A2
- (ii) $r_\beta = 25.9 \times 3.64 \times 10^3 = 7.13 \times 10^3 \text{ m}$ A1 [3]
- (d) (i) deflected upwards.....B1
but close to original directionB1
- (ii) opposite direction to α -particle and 'through side'B1 [3]
- 6 (a) greater binding energy gives rise to release of energyM1
so must be yttriumA1 [2]
- (b) probability of decayM1
of a nucleus per unit time.....A1 [2]
- (c) (i)1 $A = \lambda N$C1
 $3.7 \times 10^6 \times 365 \times 24 \times 3600 = 0.025N$C1
 $N = 4.67 \times 10^{15}$ A1 [3]
- (i)2 mass $= 0.09 \times (4.67 \times 10^{15})/(6.02 \times 10^{23})$C1
 $= 6.98 \times 10^{-10} \text{ kg}$A1 [2]
- (ii) $A = A_0 e^{-\lambda t}$
 $A/A_0 = e^{-0.025t}$ C1
 $= 0.88$A1 [2]

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GCE ADVANCED SUBSIDIARY LEVEL AND ADVANCED LEVEL

MARK SCHEME

MAXIMUM MARK: 30

SYLLABUS/COMPONENT: 9702/05

**PHYSICS
Paper 5 (Practical (A2))**



UNIVERSITY of CAMBRIDGE
Local Examinations Syndicate

Page 1	Mark Scheme	Syllabus	Paper
	A/AS LEVEL EXAMINATIONS - JUNE 2003	9702	05

- 1 (a) (v) Measurements 3
- 6 sets** of readings ($I \neq 0$) scores **1 mark**
Allow more than 6 sets without penalty
Write the number of readings as a ringed total by the table
Choose a row in the table
Check a **value for $\tan \theta$** . Tick if correct and score **1 mark**
If incorrect, write in correct value and do not award the mark
Ignore small rounding errors
All values of $\theta < 90^\circ$ score **1 mark**
Minor help from the Supervisor -1. Major help, then -2
If help has been given then write SR at the top of the front page of the script, and give a brief explanation of the type of help that has been given by the table of results
- (a) (v) Repeats 1
- Expect to see at least two sets of readings for θ , with an average calculated
Do not award this mark if all the results are the same
- (a) (v) Quality of results 2/1/0
- Judge by scatter of points about the line of best fit
6 trend points with little scatter scores 2 marks
5 trend points with little scatter scores 1 mark
Shallow curve can score 1 mark
4 trend points only scores zero
Wrong trend or 'impossible results' cannot score these marks
- (a) (v) Column headings 1
- Apply to the current column only
There must be some distinguishing mark between the quantity and the unit
Allow I/A , I (A) or I in A
- (a) (v) Consistency 2/1/0
- Apply to both θ and I
All values of θ must be given to the same number of d.p.
Allow θ to be given to the nearest half degree or nearest degree
All values of I must be given to the same number of d.p. (0.1 A or 0.01 A)
Do not accept values to the nearest Ampere or milliampere
- (a) (vi) Justification of sf in $\tan \theta$ 2/1/0
- Answer must relate the number of sf in θ to the number of sf in $\tan \theta$
Do not allow answers in terms of decimal places
'Raw data' ideas can score 1 mark
- (b) (i) Axes 1
- The axes must be labelled with the quantities plotted
Ignore units on the axes
The plotted points must occupy at least half the graph grid in both the x and y directions (i.e. 4 large squares in the x-direction and 6 large squares in the y-direction)
Do not allow more than 3 large squares between the labels on an axis
Do not allow awkward scales (e.g. 3:10, 6:10, etc.)

Page 2	Mark Scheme	Syllabus	Paper
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- (b) (i) Plotting of points 1
 All the observations must be plotted
 Count the number of plots and ring this total on the grid
 Do not allow plots in the margin area
 Check one suspect plot. Circle this plot. Tick if correct. If incorrect, mark the correct position with a small cross and use an arrow to indicate where the plot should have been, and -1. Allow errors up to and including half a small square
- (b) (i) Line of best fit 1
 Only a drawn straight line through a linear trend is allowable for this mark
 This mark can only be awarded for 5 or more plots on the grid
 There must be a reasonable balance of points about the drawn line
 Do not allow a line of thickness greater than half a small square
- (b) (ii) Gradient 1
 Ignore any units given with the value
 Hypotenuse of Δ must be $>$ half the length of line drawn
 Check the read-offs. Work to half a small square. $\Delta x/\Delta y$ gets zero
 Values taken from the table that lie on the line to within half a small square are acceptable
 Do not award this mark if a curve has been drawn
- (c) k = candidate's gradient 1
- (c) Unit of k (i.e. A^{-1}) 1
- (c) SF in k 1
 Allow 2 or 3 sf only
- (d) (i) Value of θ when $I = 15 A$ 1
 Method of working must be checked. Ignore unit and small rounding errors
- (d) (ii) Reasons for not being able to verify experimentally 1
 Heating problems with the wires
 Fuse may blow on psu/max. output current on psu exceeded
 Do not allow vague answers such as 'It is dangerous'

20 marks in total

Page 3	Mark Scheme	Syllabus	Paper
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2 A1	Sensible choice of equipment and procedure OK (i.e. measure count rate and p.d.; change p.d. and measure new count rate) Unworkable methods/inappropriate choice of apparatus cannot score this mark	1
A2	Voltmeter shown in parallel with the GM tube or the supply	1
A3	Ratemeter/scalar/datalogger connected to terminals A and B of GM tube	1
B1	Radium or Cobalt source used	1
B2	Reason for choice Answer must relate to half-life. This mark cannot be scored if B1 = 0	1
B3	Method of removing α or β radiation (depending on source used) Appropriate absorber is expected. Accept 'aluminium' or <u>thin</u> lead Could be shown on the diagram. Allow electric or magnetic deflection	1
C1/2	Any two safety precautions e.g. use source handling tool store source in lead lined box when not in use do not point source at people/do not look directly at source Do not allow 'protective clothing', 'lead suits', 'lead gloves', 'goggles', etc.	2
D1/2	Any good/further detail Examples of creditworthy points might be: Repeat readings (to allow for randomness of activity) or scalar + long time Sensible value of p.d. applied to GM tube (i.e. 50 V to 1000 V) Keep distance from source to GM tube <u>constant/fixed/same</u> , etc. <u>Subtract</u> count rate due to background radiation Aluminium sheets must be mm or cm thickness Allow other valid points. Any two, one mark each	2

10 marks in total

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GCE ADVANCED SUBSIDIARY LEVEL AND ADVANCED LEVEL

MARK SCHEME

MAXIMUM MARK: 40

SYLLABUS/COMPONENT: 9702/06

**PHYSICS
Paper 6 (Options (A2))**



UNIVERSITY of CAMBRIDGE
Local Examinations Syndicate

Page 1	Mark Scheme	Syllabus	Paper
	A/AS LEVEL EXAMINATIONS - JUNE 2003	9702	06

Categorisation of marks

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C marks: These are compensatory method marks which can be scored even if the points to which they refer are not written down by the candidate, providing subsequent working gives evidence that they must have known it. For example, if an equation carries a C-mark and the candidate does not write down the actual equation but does correct working which shows he/she knew the equation, then the C-mark is awarded.

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Conventions within the marking scheme

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Page 2	Mark Scheme	Syllabus	Paper
	A/AS LEVEL EXAMINATIONS - JUNE 2003	9702	06

Option A – Astrophysics and Cosmology

- 1 (a) large mass of gas (allow H and He) B1
giving off e.m. radiation (allow light)..... B1
held together by gravitational forces, or other good physics B1 [3]
- (b) group of (many) stars..... B1
any further detail e.g. some dimension, shape, etc B1 [2]
- (c) rocky or gaseous object B1
orbiting a star B1
seen by reflected light B1 [3]
- 2 measure wavelength of light received from galaxy B1
measure wavelength of light in laboratory/on Earth B1
(fractional) change in wavelength related to speed
or Doppler shift gives speed B1 [3]
- 3 (a) $v = H_0 d$
 $H_0 = (1.8 \times 10^4)/430$ C1
 $= 42 \text{ km s}^{-1} \text{ Mpc}^{-1}$ A1 [2]
- (b) (i) $1 \text{ pc} = 3.1 \times 10^{16} \text{ m}$ B1
age = $1/H_0$
 $= (3.1 \times 10^{22})/(42 \times 10^3)$ C1
 $= 7.4 \times 10^{17} \text{ s}$ A1
- (ii) Earth-Moon distance = $3.8 \times 10^5 \text{ km}$ (allow $2 - 7 \times 10^5 \text{ km}$)..... C1
speed = $(3.8 \times 10^8)/(7.4 \times 10^{17})$
 $= 5.1 \times 10^{-10} \text{ m s}^{-1}$ A1 [5]
- (c) This is local gravitational attraction B1
On wider scale, galaxies are receding B1 [2]

Option F – The Physics of Fluids

- 4 (a) (i) equal..... B1
- (ii) density of ice is less..... B1 [2]
- (b) mass of ice becomes equal mass of water (allow weight)..... M1
melted ice fills space of water displaced by ice M1
so level does not change A1 [3]
- 5 (a) e.g. streamline, incompressible
non-viscous, horizontal flow.....(1 each, max 3) B3 [3]
- (b) air close to train moves at the speed of the train/air dragged along
by train..... B1
air at some distance from the train is stationary/velocity is less ... B1
(so) air pressure is lower close to the train..... M1
pressure difference could force passengers into side of train A1 [4]
- 6 (a) (i) random/irregular movement (of fluid) B1
any other detail, e.g. eddies, pattern always changing..... B1

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- (ii) kinetic energy given to air to cause turbulence or work needed to overcome drag force M1
energy comes from car so fuel consumption increases A1 [4]
- (b) (i) drag coefficient/drag constant B1
- (ii) power = Fv and hence M1
 $P = \frac{1}{2}C_D\rho Av^3$ A0
- (iii) $120 \times 10^3 - \frac{1}{2} \times 0.3 \times 1.2 \times 2.5 \times v^3$ C1
 $v^3 = 2.67 \times 10^5$
 $v = 64 \text{ m s}^{-1}$ A1 [4]
- Option M – Medical Physics**
- 7 (a) electrons fired at metal target B1
electrons decelerated giving off (e.m.) radiation B1
range of decelerations, so continuous spectrum B1
also, electrons in inner orbits are excited B1
de-excitation gives characteristic line spectrum B1 [5]
- (b) (i) increase cathode/tube current B1
- (ii) increase anode voltage B1
- (iii) use aluminium filter (allow metal filter) B1 [3]
- (c) $I = I_0 e^{-\mu x}$ C1
 $\ln 2 = 0.40\mu$
 $\mu = 1.733 \text{ cm}^{-1}$ or $\ln 2 / 0.4$ C1
 $0.1 = e^{-1.733x}$
 $x = 1.33 \text{ cm}$ A1 [3]
- 8 (a) produces greater intensity (at focus)
limits region of cell damage
allows for accurate guidance B2 [2]
- (b) laser beam cauterises tissue
can produce coagulation
vaporisation of water in cells B2 [2]
{in (a) and (b), allow 1 mark each up to max of 3 in either, total not to exceed 4}
- 9 (a) ability to detect (small) changes in loudness/intensity B1
depends on $I / \Delta I$ B1 [2]
- (b) $\Delta I.L. = 10 \lg(\Delta I / I)$ or $I.L. = 10 \lg(I/I_0)$ C1
 $3.0 = 10 \lg(I_2 / (4.5 \times 10^{-5}))$ C1
 $I_2 = 9.0 \times 10^{-5} \text{ W m}^{-2}$, $\Delta I = 4.5 \times 10^{-5} \text{ W m}^{-2}$ A1 [3]

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Option P – Environmental Physics

- 10 (a) source of (useful) energy B1
derived from (incomplete) decay of organic matter B1 [2]
- (b) resources: total deposits of fossil fuels B1
reserves: fossil fuels that can be extracted (economically) B1 [2]
- 11 (a) heavy nucleus/heavy atom/U-235, etc B1
bombarded by neutron B1
produces two fragments of about equal mass B1
plus neutrons and energy B1 [4]
- (b) (i) slows down neutrons B1
(ii) absorbs neutrons B1
(iii) maintains coolant around reactor core B1
provides biological shield/prevents radiation leakage B1 [4]
- 12 (a) $E_{\text{MAX}} = (1 - T_L/T_H)$ C1
 $= (1 - 313/813)$ C1
 $= 0.61$ A1 [3]
- (b) (i) e.g. heat loss in exhaust gases/cooling towers B1
(ii) e.g. pre-heat water entering boiler, either increase T_H or decrease T_L
re-heat steam in multistage turbine, CHP system...(1 each, max 2) B2 [3]
- (c) e.g. thermal, visual, etc.....(1 each, max 2)..... B2 [2]

Option T – Telecommunications

- 13 (a) correct signal voltages.....(-1 each error or omission) B2
corresponding binary numbers...(-1 each error or omission)..... B2 [4]
- (b) signal changes at correct positions B1
correct levels B1 [2]
- (c) (use ADC and DAC with) larger number of bits M1
makes smaller 'step height' A1
sample more frequently M1
makes smaller 'step depth' A1 [4]
- 14 (a) central conductor with outer screening B1
insulation between inner and outer and also as cladding B1 [2]
- (b) e.g. greater bandwidth
immune to e.m. interference
radiates less e.m. power
less cross-talk
lower noise levels..... (1 each, max 3)..... B3 [3]
- 15 10 m → 100 m worldwide
more than 100 m 1000 km
less than 10 m line of sight or worldwide using satellites
(-1 each error or omission) B5 [5]